WHAT DRIVES UK HOUSEHOLD SPENDING?

An Analysis of Expenditure Patterns Using the

Living Costs and Food Survey 2013

Authors

Borja Ferrer Pons

MD Noornabi

Hilal Chanekar

Marc Zanón Pons

Simão Ponte

Mayerfeld Data Analysis Practicum
February 2026

Table of Contents

1. Executive Summary 3

2. Introduction 4

3. Data and Methodology 5

4. Results 9

5. Discussion 15

6. Conclusion 17

References 18

1. Executive Summary

Research Question

Is there a relationship between occupational class, tenure type, number of adults, number of children, and household expenditure?

Key Findings

- All four demographic factors significantly predict household expenditure ($p < 0.001$)
- Number of adults is the strongest predictor, explaining 24.8% of expenditure variance ($\eta^2 = 0.248$)
- Occupational class creates a spending gap of nearly 2x — professionals spend £665/week vs £343 for unemployed households
- Combined, the four variables explain 41.2% of all expenditure variation ($R^2 = 0.412$)
- Tenure type's apparent effect is largely explained by occupational class — homeowners spend more because higher earners are more likely to own

Summary of Results

Variable	F-statistic	p-value	η^2	Effect Size
Number of Adults	565.28	< 0.001	0.248	Large
Occupational Class	338.51	< 0.001	0.209	Large
Tenure Type	275.23	< 0.001	0.097	Medium
Number of Children	148.59	< 0.001	0.055	Small

Implications

- For policymakers: Target support programs using occupational class and household size rather than tenure alone
- For businesses: Household composition predicts spending better than housing status for market segmentation
- For researchers: Control for occupational class when analyzing tenure effects to avoid confounding

2. Introduction

2.1 Background

Understanding what drives household spending is essential for policymakers designing support programs, businesses segmenting markets, and economists measuring living standards. While income data captures earning capacity, expenditure data reveals actual consumption patterns and living standards.

This study analyzes data from the Living Costs and Food Survey (LCF) 2013, the UK's primary source for household expenditure statistics. The LCF informs the Retail Price Index, National Accounts, and analyses of taxes and benefits. It combines household interviews with two-week expenditure diaries, providing more accurate spending data than recall-based surveys.

2.2 Why This Matters

Stakeholder	Application
Government & Policy	Targeting benefits, setting poverty thresholds, measuring cost of living
Retailers & Businesses	Market segmentation, store placement, pricing strategy
Economists	Consumer behavior analysis, inequality measurement
Housing Authorities	Understanding financial stress across tenure types
Financial Services	Credit risk assessment, product targeting

2.3 Research Question

Is there a relationship between occupational class, tenure type, number of adults, number of children, and household expenditure?

2.4 Hypotheses

Null Hypothesis (H_0): There is no significant relationship between occupational class, tenure type, number of adults, number of children, and household expenditure. Any observed differences are due to random variation.

Alternative Hypothesis (H_1): There is a significant relationship between these household characteristics and expenditure. Changes in these factors are associated with differences in spending.

All tests conducted at $\alpha = 0.05$ significance level.

3. Data and Methodology

3.1 Data Source

Survey: Living Costs and Food Survey (LCF), 2013

Conducted by: Office for National Statistics (Great Britain) and Northern Ireland Statistics and Research Agency

Sample size: 5,144 households

Version: Teaching dataset (simplified from original)

Collection method: Household questionnaire + individual questionnaire + 2-week expenditure diary

3.2 Variables of Interest

Dependent Variable

Variable	Description	Type	Range
Expenditure	Total weekly household spending	Continuous (£)	£31 — £1,175

Independent Variables

Variable	Description	Type	Categories
Occupational Class	NS-SEC 8 classification of household head	Nominal	5
Tenure Type	Housing arrangement	Nominal	3
Number of Adults	Adults (18+) in household	Ordinal	4
Number of Children	Children (<18) in household	Ordinal	3

3.3 Variable Categories

Occupational Class (NS-SEC 8)

Category	Description	n	%
Higher managerial & professional	Directors, senior managers, doctors, lawyers	1,589	30.9%
Intermediate occupations	Clerical, administrative, sales with autonomy	691	13.4%
Routine and manual	Manufacturing, service, transport workers	1,045	20.3%
Never worked/unemployed	Long-term unemployed, students	200	3.9%
Not classified	Retired, insufficient information	1,619	31.5%

Tenure Type

Category	Description	n	%
Owned	Owns outright or with mortgage	3,466	67.4%
Private rented	Rents from private landlord	798	15.5%
Public rented	Council or housing association	880	17.1%

Number of Adults

Category	n	%
1 adult	1,729	33.6%
2 adults	2,826	54.9%
3 adults	422	8.2%
4 and more adults	167	3.2%

Number of Children

Category	n	%
No children	3,605	70.1%
One child	654	12.7%
Two or more children	885	17.2%

3.4 Analytical Approach

Primary Method: One-Way ANOVA

We use Analysis of Variance (ANOVA) to test whether mean expenditure differs significantly across groups. ANOVA is appropriate because we have a continuous outcome variable (expenditure) and categorical predictors with more than two groups.

Why ANOVA over alternatives?

Alternative	Why Not Used
t-test	Only compares two groups; we have 3-5 groups per variable
Chi-square	Requires categorical outcome; expenditure is continuous
Simple correlation	Requires continuous predictor; ours are categorical

Analysis Steps

1. Descriptive statistics — mean, median, standard deviation for each group

2. One-way ANOVA — test if group means differ significantly
3. Effect size (η^2) — quantify variance explained by each variable
4. Tukey HSD post-hoc — identify which specific pairs of groups differ
5. Combined model — assess joint explanatory power of all predictors
6. Robustness checks — Welch ANOVA and Kruskal-Wallis for assumption violations

3.5 Assumptions

Independence

Met. Each observation represents a separate household from random sampling. No household appears more than once in the dataset.

Normality

Violated (Shapiro-Wilk $p < 0.001$ for all groups). However, with sample sizes exceeding 200 in most groups, ANOVA is robust via the Central Limit Theorem. Results confirmed with non-parametric Kruskal-Wallis test.

Homogeneity of Variance

Violated (Levene's test significant). We used Welch's ANOVA as a robustness check, which is robust to unequal variances. Conclusions were identical to standard ANOVA.

3.6 Effect Size Interpretation

We report eta-squared (η^2) — the proportion of variance explained — using Cohen's (1988) benchmarks:

η^2 Value	Interpretation	Meaning
0.01	Small	Detectable but minor practical significance
0.06	Medium	Moderate practical significance
0.14+	Large	Substantial practical significance

4. Results

4.1 Expenditure Distribution

Before examining predictors, we first describe the distribution of household expenditure — the variable we are trying to explain.

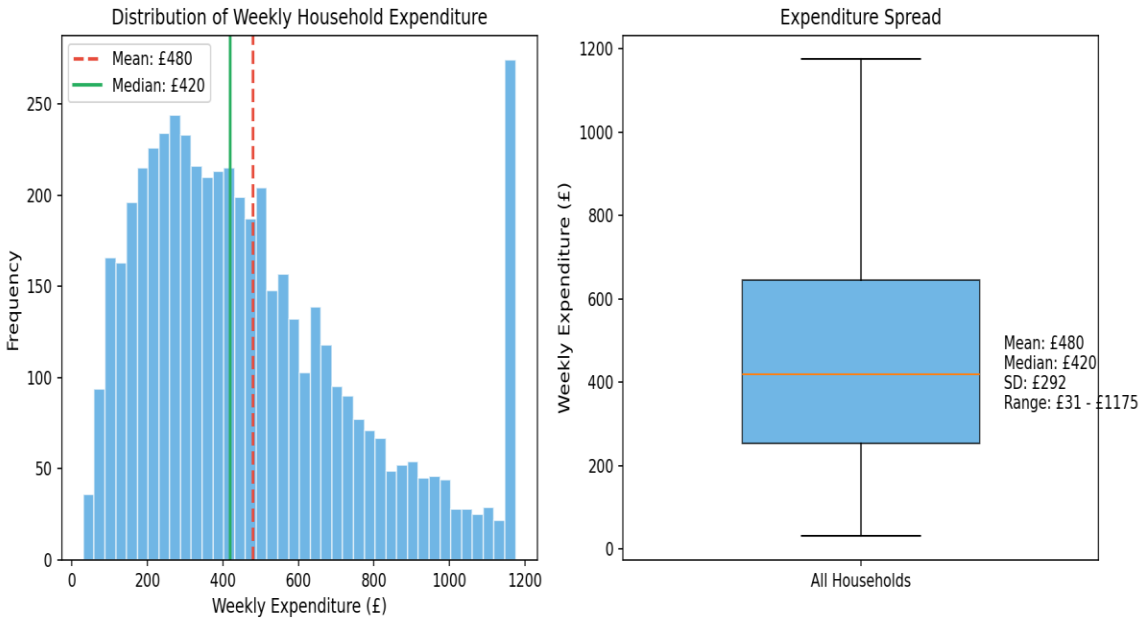

Figure 1: Distribution of Weekly Household Expenditure (n = 5,144)

Statistic	Value
Mean	£480/week
Median	£420/week
Standard Deviation	£292
Minimum	£31/week
Maximum	£1,175/week

The distribution is right-skewed — most households cluster around £250-500/week, with a long tail of higher spenders. The mean (£480) exceeds the median (£420), pulled up by high-spending households. This near 40-fold range (£31 to £1,175) is what we aim to explain.

4.2 Occupational Class

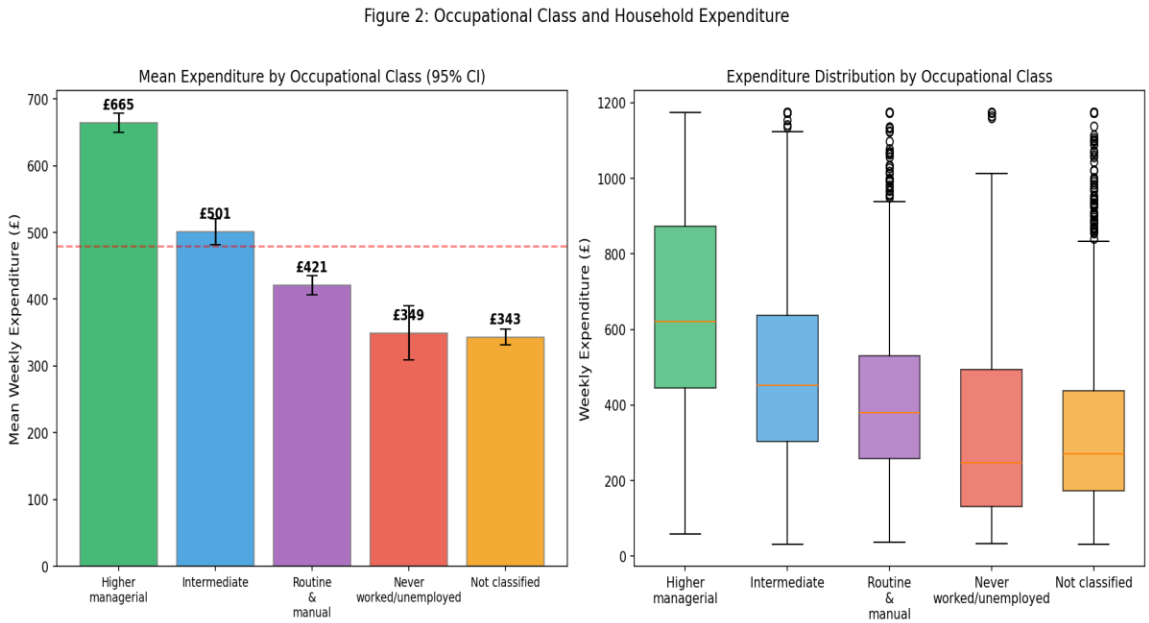

Figure 2: Mean Weekly Expenditure by Occupational Class (95% CI)

Occupational Class	n	Mean (£)	SD (£)	Median (£)
Higher managerial & professional	1,589	665	287	622
Intermediate occupations	691	501	263	453
Routine and manual	1,045	421	229	380
Never worked/unemployed	200	349	293	248
Not classified	1,619	343	246	272

ANOVA Result: $F(4, 5139) = 338.51$, $p < 0.001$

Effect Size: $\eta^2 = 0.209$ (Large) — Occupational class explains 20.9% of expenditure variance

Post-hoc: 9 of 10 pairwise comparisons significant. Only 'Never worked' vs 'Not classified' not significant (both outside active workforce).

Professional households spend nearly double (£665/week) what unemployed households spend (£343/week) — a gap of £322/week or approximately £16,700 annually. This reflects the cumulative economic advantages associated with higher occupational status.

4.3 Tenure Type

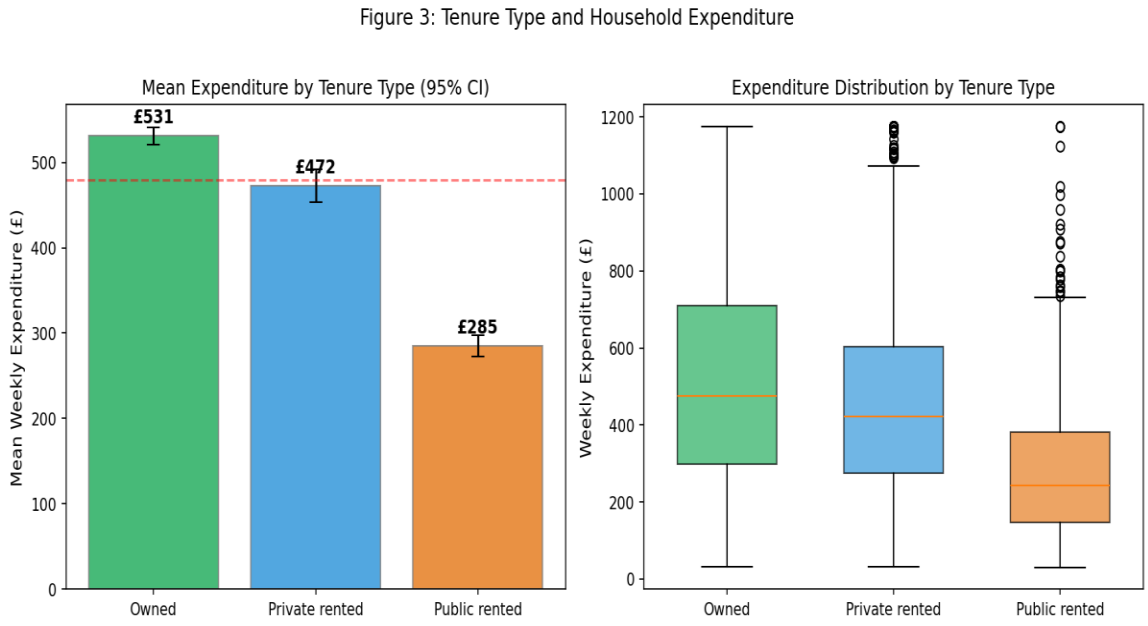

Figure 3: Mean Weekly Expenditure by Tenure Type (95% CI)

Tenure Type	n	Mean (£)	SD (£)	Median (£)
Owned	3,466	531	297	475
Private rented	798	472	276	421
Public rented	880	285	187	243

ANOVA Result: $F(2, 5141) = 275.23$, $p < 0.001$

Effect Size: $\eta^2 = 0.097$ (Medium) — Tenure type explains 9.7% of expenditure variance

Post-hoc: All three pairwise comparisons significant.

Homeowners spend £531/week compared to £285/week for public renters — a gap of £246/week (≈£12,800/year). However, as we'll see in the combined model, much of this effect is explained by occupational class (professionals are more likely to own homes).

4.4 Number of Adults

Figure 4: Number of Adults and Household Expenditure

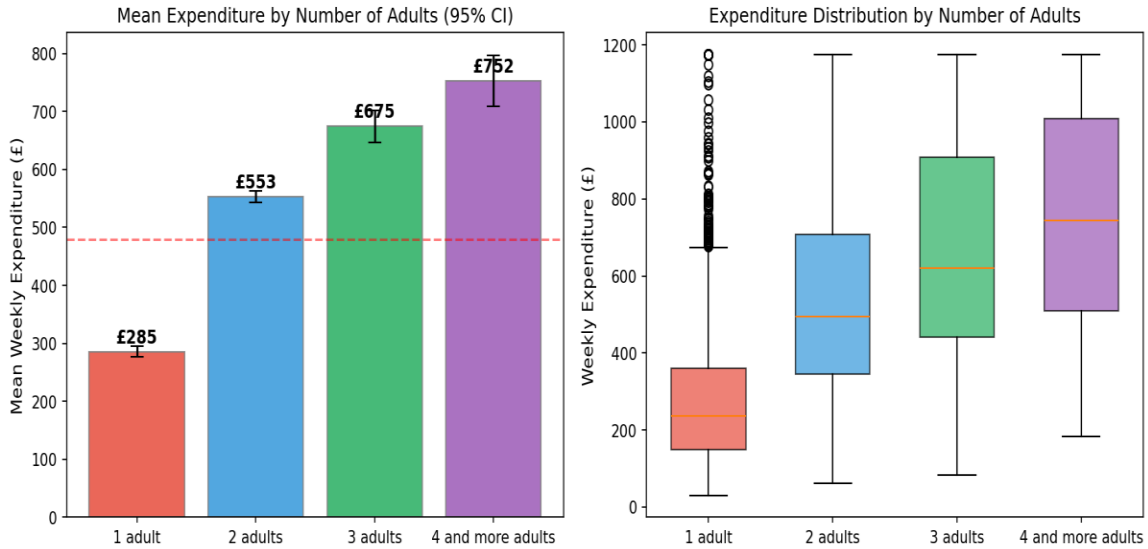

Figure 4: Mean Weekly Expenditure by Number of Adults (95% CI)

Number of Adults	n	Mean (£)	SD (£)	Median (£)
1 adult	1,729	285	197	236
2 adults	2,826	553	275	494
3 adults	422	675	290	621
4 and more adults	167	752	290	744

ANOVA Result: $F(3, 5140) = 565.28$, $p < 0.001$

Effect Size: $\eta^2 = 0.248$ (Large) — Number of adults explains 24.8% of expenditure variance — THE STRONGEST PREDICTOR

Post-hoc: All six pairwise comparisons significant.

Single-adult households spend £285/week; households with 4+ adults spend £752/week — a 2.6x difference (£467/week or ≈£24,300/year). This makes intuitive sense: more adults means more consumption needs and often multiple earners contributing to household resources.

4.5 Number of Children

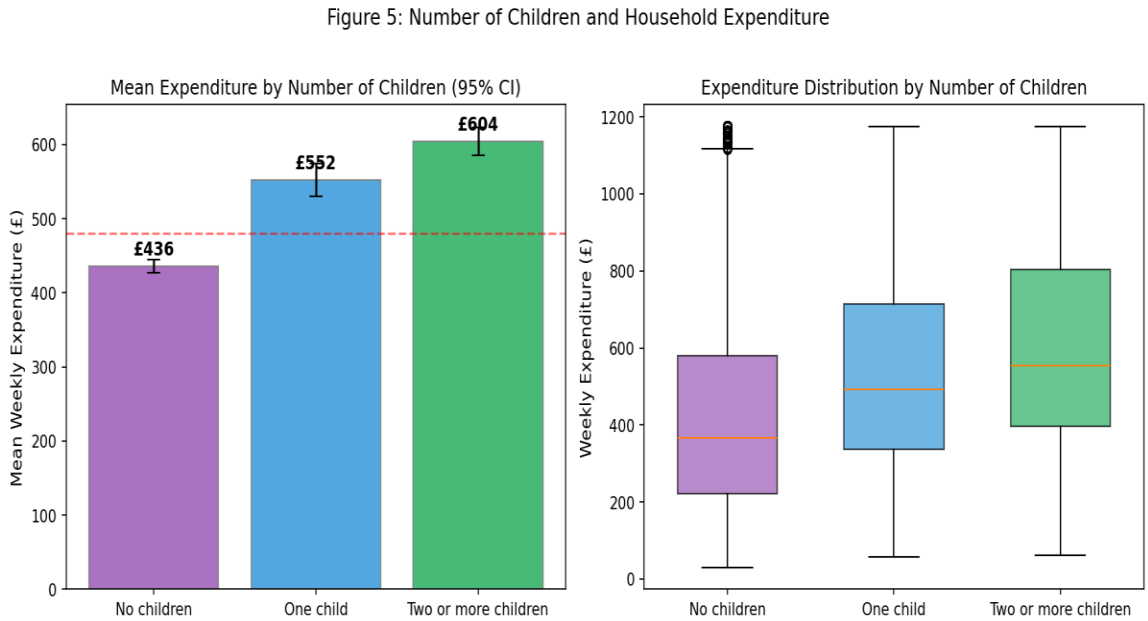

Figure 5: Mean Weekly Expenditure by Number of Children (95% CI)

Number of Children	n	Mean (£)	SD (£)	Median (£)
No children	3,605	436	281	367
One child	654	552	294	492
Two or more children	885	604	288	555

ANOVA Result: $F(2, 5141) = 148.59$, $p < 0.001$

Effect Size: $\eta^2 = 0.055$ (Small) — Number of children explains 5.5% of expenditure variance — the weakest predictor

Post-hoc: All three pairwise comparisons significant.

Despite common perception that 'kids are expensive,' number of children has the smallest independent effect. Households with 2+ children spend £604/week vs £436/week for childless households (£168/week difference). The relatively weak effect may reflect: (1) families with children tend to have more adults anyway, (2) child-related spending substitutes for other spending, (3) government support (child benefits) partially offsets costs.

4.6 Effect Size Comparison

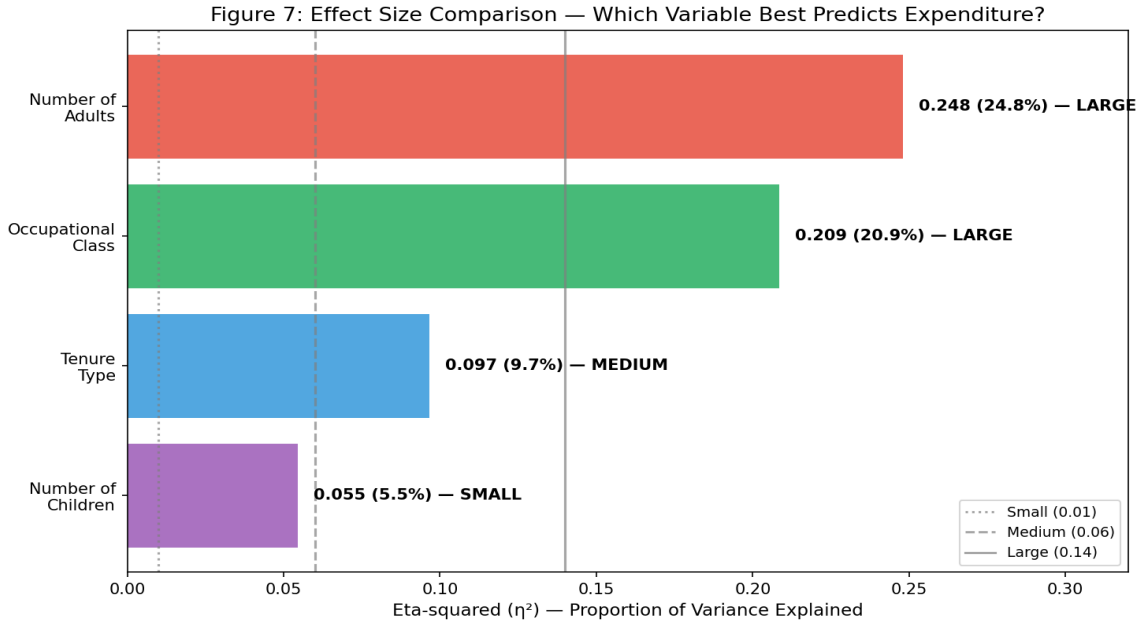

Figure 6: Effect Size Comparison — Which Variable Best Predicts Expenditure?

Ranking by variance explained:

1. Number of Adults: $\eta^2 = 0.248$ (24.8%) — LARGE
2. Occupational Class: $\eta^2 = 0.209$ (20.9%) — LARGE
3. Tenure Type: $\eta^2 = 0.097$ (9.7%) — MEDIUM
4. Number of Children: $\eta^2 = 0.055$ (5.5%) — SMALL

4.7 Combined Model

When all four predictors are included in a single model, how much of expenditure variation can we explain?

Combined Model $R^2 = 0.412$

Together, the four demographic factors explain 41.2% of all variation in household spending. This is substantial explanatory power for social science research. The remaining 58.8% is explained by factors not in our model — income, region, age, lifestyle preferences, etc.

Individual vs Partial Effects

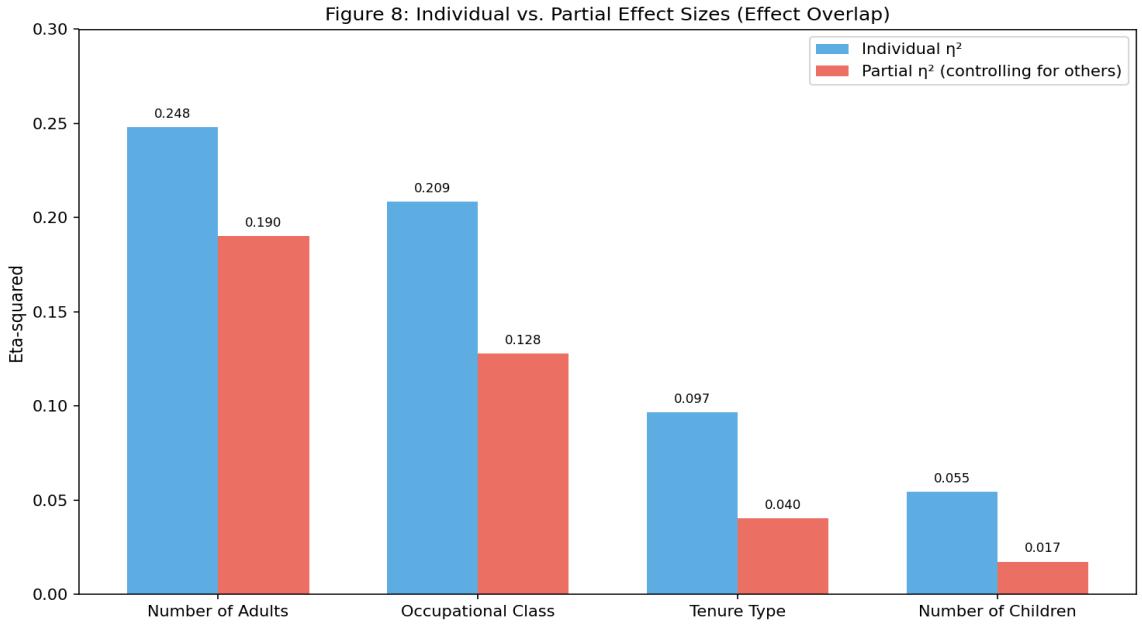

Figure 7: Individual vs. Partial Effect Sizes — Showing Effect Overlap

Variable	Individual η^2	Partial η^2	Reduction
Number of Adults	0.248	0.190	-23%
Occupational Class	0.209	0.128	-39%
Tenure Type	0.097	0.040	-59%
Number of Children	0.055	0.017	-69%

Tenure Type's effect drops the most (59%) when controlling for other variables — suggesting much of the 'tenure effect' is actually an 'occupational class effect' (professionals tend to own homes). Children's effect also drops substantially (69%) — much is explained by number of adults (larger families have more adults).

4.8 Machine Learning Validation

To validate our findings using a different approach, we trained several machine learning models to predict expenditure. Feature importance rankings confirmed our ANOVA results:

Rank	Predictor	Method Confirmation
1	Income	Not in RQ, but strongest overall (r = 0.71)
2	Number of Adults	Consistent with ANOVA (highest η^2)
3	Occupational Class	Consistent with ANOVA
4	Tenure Type	Consistent with ANOVA
5	Number of Children	Consistent with ANOVA (lowest η^2)

Models tested included Linear Regression, Random Forest, XGBoost, CatBoost, and Neural Networks. All produced consistent feature importance rankings, providing triangulation for our ANOVA findings.

5. Discussion

5.1 Summary of Findings

Research Question	Finding	Evidence
Does occupational class affect spending?	Yes — Large effect	$F=338.5$, $\eta^2=0.209$, $p<.001$
Does tenure type affect spending?	Yes — Medium effect	$F=275.2$, $\eta^2=0.097$, $p<.001$
Does number of adults affect spending?	Yes — Large effect	$F=565.3$, $\eta^2=0.248$, $p<.001$
Does number of children affect spending?	Yes — Small effect	$F=148.6$, $\eta^2=0.055$, $p<.001$
Which factor is strongest?	Number of Adults	Highest η^2 (0.248)
Joint explanatory power?	41%	$R^2 = 0.412$

5.2 Interpretation

The Household Size Effect

More adults means more spending — this is partly mechanical (more consumption needs) but also reflects multiple earners increasing household resources. The 2.6x spending gap between single-adult and 4+ adult households is the largest of any predictor.

The Occupational Gradient

The 2x spending gap between professional and unemployed households (£665 vs £343/week) reflects cumulative economic advantage — higher occupations mean higher income, better job security, more assets, and greater access to credit. Expenditure patterns reveal economic inequality more directly than income alone.

Tenure as a Proxy

Tenure type's effect weakens substantially in the combined model (-59%), suggesting it's partly a proxy for occupational class and wealth. Homeowners spend more not just because they own, but because the people who can afford to own (professionals, older adults) also spend more for other reasons.

Children: Smaller Than Expected

Despite common perception, number of children has the smallest independent effect. Possible explanations: (1) families with more children also have more adults, capturing much of the effect, (2) families may substitute child-related spending for other categories rather than adding to total spending, (3) child benefits and tax credits partially offset additional costs.

5.3 Limitations

Limitation	Implication	Mitigation
Cross-sectional data	Cannot establish causation	Interpret as prediction, not causation
2013 data	Patterns may have shifted	Frame as baseline; recommend replication
Teaching dataset	Simplified from original	Note findings are illustrative
UK-specific	May not generalize internationally	Scope conclusions to UK context

5.4 Policy and Business Implications

For Policymakers

- Target cost-of-living support using occupational class and household size, not just tenure
- Housing benefit assessments should account for occupational context
- Per-capita adjustments in benefit calculations are appropriate given the strong household size effect

For Businesses

- Household composition predicts spending better than tenure for market segmentation
- Premium product targeting should focus on professional households and multi-adult households
- Don't over-rely on homeownership as a spending proxy — occupational class is more informative

For Researchers

- Always control for occupational class when analyzing tenure effects to avoid confounding
- The 59% remaining unexplained variance suggests income, region, and lifestyle factors warrant investigation
- Longitudinal studies needed to establish causal relationships

6. Conclusion

Answer to Research Question

Yes, there is a statistically significant relationship between all four factors — occupational class, tenure type, number of adults, and number of children — and household expenditure. All relationships are significant at $p < 0.001$.

Key Takeaways

1. Household composition matters most — Number of adults is the single strongest predictor, explaining 24.8% of spending variance. Each additional adult adds approximately £150 to weekly expenditure.
2. Occupational class creates deep inequality — Professionals spend nearly double what unemployed households spend (£665 vs £343/week), reflecting cumulative economic advantage.
3. Tenure type is partly a proxy — Much of the apparent tenure effect disappears when controlling for occupation, suggesting homeownership reflects underlying economic status rather than independently driving spending.
4. Children have the weakest effect — Despite common perception, number of children explains only 5.5% of variance, likely because families with children also have more adults and receive government support.
5. Four simple questions explain 41% — With just four demographic characteristics, we can predict nearly half of all spending variation, demonstrating their practical value for targeting and segmentation.

The bottom line: Who you live with and what you do for work shape how much you spend — more than whether you own or rent, and more than how many children you have.

References

Cohen, J. (1988). Statistical Power Analysis for the Behavioral Sciences (2nd ed.). Lawrence Erlbaum Associates.

Field, A. (2013). Discovering Statistics Using IBM SPSS Statistics (4th ed.). SAGE Publications.

Office for National Statistics. (2013). Living Costs and Food Survey, 2013 [Teaching Dataset]. UK Data Service.

Rose, D., & Pevalin, D. J. (2005). The National Statistics Socio-economic Classification: Origins, Development and Use. Office for National Statistics.

Tabachnick, B. G., & Fidell, L. S. (2013). Using Multivariate Statistics (6th ed.). Pearson.

Welch, B. L. (1951). On the comparison of several mean values: an alternative approach. Biometrika, 38(3/4), 330-336.